DR. ARIS THORNE

aris.thorne@quantumai.org | +1-617-555-0199 | Cambridge, MA | linkedin.com/in/aristhorne

EXECUTIVE SUMMARY

EXECUTIVE & RESEARCH EXPERIENCE

Quantum AI Labs · Vice President of Research & Engineering

Cambridge, MA | 2020 · Present

? Delivered next-generation reasoning orchestration microservices adopted by 4.2M enterprise developer accounts globally.

? Negotiated multi-year strategic research alliances with leading academic institutions and national research laboratories.

NeuralEdge Systems · Senior Director of Distributed Systems

Boston, MA | 2016 · 2020

? Architected real-time edge intelligence inference pipelines deployed across 450,000 industrial IoT sensor gateways.

PRIOR INDUSTRY EXPERIENCE (CONTINUED)

Apex Distributed Networks · Lead Infrastructure Architect

New York, NY | 2011 · 2016

? Pioneered petabyte-scale stream processing framework utilizing Apache Kafka and custom zero-copy memory buffers.

Vanguard Cybernetics · Principal Software Engineer

Waltham, MA | 2006 · 2011

? Designed high-assurance cryptographic kernel drivers for mission-critical defense and aerospace telemetry platforms.

? Formalized static verification test harnesses reducing firmware defect density by 64% prior to silicon tape-out.

Solaria Information Systems · Senior Software Engineer

Cambridge, MA | 2002 · 2006

? Engineered transactional relational database storage engines handling distributed two-phase commit consensus algorithms.

? Authored B-tree indexing and Write-Ahead Log recovery components supporting high-concurrency ACID transactions.

EDUCATION & ACADEMIC CREDENTIALS

Massachusetts Institute of Technology (MIT)

Ph.D. in Computer Science & Artificial Intelligence | 1998 · 2002

? Dissertation: Asynchronous Distributed Consensus Protocols in Heterogeneous Byzantine Environments.

? Advisor: Prof. Leslie M. Sterling | NSF Graduate Research Fellow.

Stanford University

Master of Science in Computer Science | 1996 · 1998

? Focus: Distributed Operating Systems, Compilers, and Algorithmic Complexity | GPA: 4.0 / 4.0

Cornell University

Bachelor of Science in Electrical Engineering & Computer Science | 1992 · 1996

? Summa Cum Laude | Tau Beta Pi Engineering Honor Society | Dean's Honor List all semesters.

COMPREHENSIVE TECHNICAL COMPETENCIES

? AI & Deep Learning: PyTorch, TensorFlow, JAX, CUDA, TensorRT, Triton, DeepSpeed, HuggingFace, ONNX, vLLM

? Languages: Python, C++, C, Rust, Go, CUDA C, SQL, Assembly x86_64, Julia, Shell Scripting

? Distributed Systems: Kubernetes, Apache Spark, Kafka, Ray, Slurm, Redis, Cassandra, Etcd, gRPC, Protobuf

? Cloud & Hardware: AWS, GCP, Azure, NVIDIA H100/A100 Clusters, InfiniBand HDR/NDR, RoCE v2, Docker

? Leadership & Governance: Research Program Management, Technology Roadmap Strategy, P&L Oversight, IP Strategy

SELECTED PEER-REVIEWED PUBLICATIONS

[1] Thorne, A., & Mercer, K. (2024). Scalable Speculative Decoding Across Decentralized Inference Meshes. In Proceedings of NeurIPS.

[2] Thorne, A., et al. (2023). Adaptive Gradient Sparsification in Massive-Scale Multi-GPU Neural Networks. In Proceedings of ICML.

[3] Vance, D., & Thorne, A. (2022). Resilient Zero-Knowledge Consensus Protocols for Byzantine Distributed Ledgers. ACM Transactions on Computer Systems (TOCS).

[4] Thorne, A., & Sterling, L. (2020). Fault-Tolerant Dynamic Load Rebalancing in High-Throughput Stream Processing. IEEE Transactions on Cloud Computing.

[5] Thorne, A., & Zhang, W. (2018). Sub-Millisecond Telemetry Serialization for Edge Industrial Sensor Grids. In Proceedings of ACM SIGCOMM.

[6] Thorne, A. (2016). Formal Verification of Real-Time Cryptographic Primitives in Microkernel Architectures. In Proceedings of USENIX OSDI.

[7] Morales, C., & Thorne, A. (2014). High-Concurrency Lock-Free Queue Structures for Cache-Coherent Multi-Core Processors. ACM SIGPLAN PPoPP.

[8] Thorne, A., & Patel, M. (2012). Petabyte-Scale Stream Ingestion: Lessons from Production Deployments. In Proceedings of VLDB.

[9] Thorne, A., & Sterling, L. (2002). Asynchronous Consensus Bounds in Partially Synchronous Networks. Journal of the ACM (JACM).

[10] Thorne, A. (1998). Dynamic Thread Scheduling in Multiprocessor Unix Kernels. IEEE Micro.

ISSUED PATENTS

? US Patent 11,842,109: Distributed Parallel Tensor Computation with Adaptive Memory Swapping (Issued 2023).

? US Patent 10,951,682: High-Speed Zero-Copy Stream Ingestion in Multi-Node Clusters (Issued 2021).

? US Patent 10,218,443: Autonomous Network Latency Mitigation in Edge Telemetry Systems (Issued 2019).

? US Patent 9,641,502: Lock-Free Buffer Serialization for Shared-Memory Multiprocessor Architectures (Issued 2017).

? US Patent 8,914,339: Fault-Tolerant Cryptographic Verification for Microkernel Telemetry Platforms (Issued 2014).

KEY INITIATIVES & STRATEGIC PROJECTS

Project TensorWeave · Technologies: Python, C++, CUDA, Triton

? Distributed compiler framework optimizing heterogeneous multi-cloud neural training jobs across thousands of GPUs.

Project ChronosMesh · Technologies: Rust, Raft, gRPC, DPDK

? High-frequency distributed consensus ledger providing microsecond-level synchronization across transatlantic financial nodes.

Project EdgeSentry · Technologies: C, Assembly, TensorFlow Lite

? Ultra-low power autonomous anomaly classification inference agent deployed across aerospace embedded microcontrollers.

HONORS & PROFESSIONAL FELLOWSHIPS

? Fellow, Association for Computing Machinery (ACM) ? Elected for contributions to distributed computing systems (2021).

? Fellow, Institute of Electrical and Electronics Engineers (IEEE) ? Elected for leadership in scalable neural architectures (2019).

? National Science Foundation (NSF) Graduate Research Fellowship Award Winner (1998).